



Antigravity DSA Visualizer

Premium AI-Powered Sandbox & Commentary

Import from LeetCode (Optional):

https://leetcode.com/problems/ ...

Fetch

Or choose a Template:

Top K Elements (Priority Heap)



Batch test cases (one input per line, optional):

4, 2, 5
1, 2, 3



Test input (array, grid, or graph edges):

[1, 2, 2, 3, 3, 3, 4, 5]

C++ Solution (use stack/queue/deque + SDK helpers):

```
1 class Solution {
2     public:
3         vector<int> topK(vector<int>& nums) {
```

```
5      for (int i = 0; i < (int)nums.size(); i++) {
6          pq.push(i);
7      }
8      vector<int> res;
9      while (!pq.empty()) {
10         int idx = pq.top();
11         pq.pop();
12         res.push_back(nums[idx]);
13     }
14     return res;
15 }
16 };
```

☐ Skip AI commentary (deterministic messages only)

⚡ Auto-Instrument Code

Generate Trace & AI Commentary

Run all cases

Load demo trace

💡 AI Suggested Edge Cases

Single-Element Array

This test case ensures that the algorithm correctly handles an array with only one element. Since there is no duplicate element to consider, it should simply return the single element.

Load Case

Mixed-Length Arrays

This test case checks how the algorithm handles arrays of different lengths. The first array has only one element, while the second array has two elements. This edge case helps verify that the algorithm can correctly handle this scenario.

Load Case

Negative and Positive Elements

This test case ensures that the algorithm works correctly with both negative and positive numbers. The presence of a negative number in the first array adds an extra layer of complexity to verify.

Load Case

Input Array

Priority Heap

AI Complexity Analysis

TIME COMPLEXITY

$O(N \log N)$

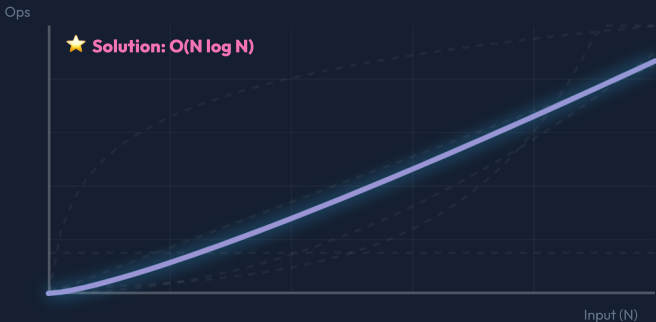
SPACE COMPLEXITY

$O(N)$

Bottlenecks & Dynamic Allocations

The primary bottleneck is the construction of the priority queue, which takes $O(N)$ time. The sorting operation within the priority queue itself also contributes to the logarithmic growth.

Interactive Growth Curve



3

[4]

3

[5]

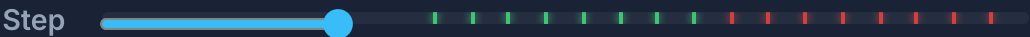
4

[6]

5

[7]

Algorithm finished.



► Step list

← Prev

Step: 26 / 26

Next →

► Auto-play

Speed: Normal (1s) ▼

Download JSON

